

Engineering Mathematics: Cayley-Hamilton Theorem

This note focuses on the verification of the Cayley-Hamilton Theorem and the subsequent calculation of a matrix inverse.

1. Theoretical Foundation

The Cayley-Hamilton Theorem

The Cayley-Hamilton Theorem states that every square matrix satisfies its own **characteristic equation**. If the characteristic polynomial of an $n \times n$ matrix A is given by $P(\lambda) = \det(A - \lambda I)$, then:

$$P(A) = 0$$

where 0 is the null matrix of order n .

Conceptual Logic

- Characteristic Equation:** Find the eigenvalues by solving $\det(A - \lambda I) = 0$.
- Substitution:** Replace the scalar variable λ with the matrix A , and replace the constant term c with cI (where I is the Identity matrix).
- Verification:** Perform matrix arithmetic to prove the resulting expression equals the zero matrix.
- Inverse Calculation:** Multiply the characteristic equation by A^{-1} to isolate and solve for the inverse matrix.

2. Step-by-Step Verification & Application

Matrix given:

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

Step 1: Find the Characteristic Equation

We solve $\det(A - \lambda I) = 0$:

$$\det \left(\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 0$$
$$\begin{vmatrix} 1 - \lambda & 2 \\ -1 & 3 - \lambda \end{vmatrix} = 0$$

Expanding the determinant:

$$(1 - \lambda)(3 - \lambda) - (2)(-1) = 0$$
$$3 - \lambda - 3\lambda + \lambda^2 + 2 = 0$$
$$\lambda^2 - 4\lambda + 5 = 0$$

Step 2: Verify Cayley-Hamilton Theorem

According to the theorem, $A^2 - 4A + 5I = 0$.

Calculate A^2 :

$$A^2 = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} (1)(1) + (2)(-1) & (1)(2) + (2)(3) \\ (-1)(1) + (3)(-1) & (-1)(2) + (3)(3) \end{bmatrix} = \begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix}$$

Substitute into the equation:

$$\begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} + 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ -4 & 12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\begin{bmatrix} -1 - 4 + 5 & 8 - 8 + 0 \\ -4 - (-4) + 0 & 7 - 12 + 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The theorem is verified.

Step 3: Find the Inverse Matrix (A^{-1})

Starting from the verified equation:

$$A^2 - 4A + 5I = 0$$

Multiply the entire equation by A^{-1} :

$$A^{-1}(A^2) - 4A^{-1}(A) + 5A^{-1}(I) = 0$$

$$A - 4I + 5A^{-1} = 0$$

Isolate $5A^{-1}$:

$$5A^{-1} = 4I - A$$

$$5A^{-1} = 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$5A^{-1} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \longrightarrow 5A^{-1} = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$$

Final result for A^{-1} :

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$$